

APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CONTINUOUS TEST 1 (CT-1) EXAMINATION • AY 2024-25

Course: Database Management Systems (CS501) | Faculty: Dr. Anil Sharma

Date: 25 September 2024

Time: 60 Minutes

Max Marks: 20

Instructions: Answer all questions in Part A. Part B contains internal choice.

PART A — FORMAL RELATIONAL THEORY

Answer all questions.

1. Define Tuple Relational Calculus (TRC) and write a TRC expression to find employees who work on all projects. [10 M]
2. Explain lossless-join decomposition and dependency preservation theorems with mathematical proofs. [10 M]